Demonstrating the physics of metamaterials with acoustic waves and soda cans

(Progress Report)

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Objective

1. To study physics of a soda can as a Helmholtz resonator by measuring the amplitude and phase response of the soda-can resonator as a function of the driving frequency.
2. To investigate the resonance frequency of the soda can for the different internal volumes.
3. To construct and characterize an acoustic antenna capable of redirecting sound preferentially in one direction.

Summary of Work Carried Out

Week 1

- During the first week, I mainly focused on understanding the theoretical background of the experiment and tried to get the mathematical concepts of Helmholtz resonator to get started with the first part of the experiment.
- Then, we tried to collect the necessary equipment needed for the experiment. One issue we faced with installation of drivers for microphone and loudspeaker.

Week 2

- A PicoScope was used as an alternative to the conventional oscilloscope and function generator, as it fulfills the functions of both instruments.
- After this, `PicoScope 7 T&M` software was installed. We read its official manuals to get familiarized with the software.
- Initially, we took data for the resonance condition manually for the empty soda can and with 200 ml filled water soda can. Although the data was not clean (noise there) but we were able to observe the resonance around frequency 300Hz for empty can.

Week 3

- Since, it was not an efficient way to collect data manually. So, we switched to C-based python wrapper '`picosdk`' to collect data quickly with plots. In this way, we were able to control PicoScope directly through the python code.

- We did changes (using some online stuffs, GitHub and AI) in the code to get the desired results.
- One of the collected data for empty soda can is shown in the fig. 1. From the figure, we can see resonance plot is quite satisfactory but further work is required to improve the phase measurement. Also, significant background and instrumental noise was present in the plots.

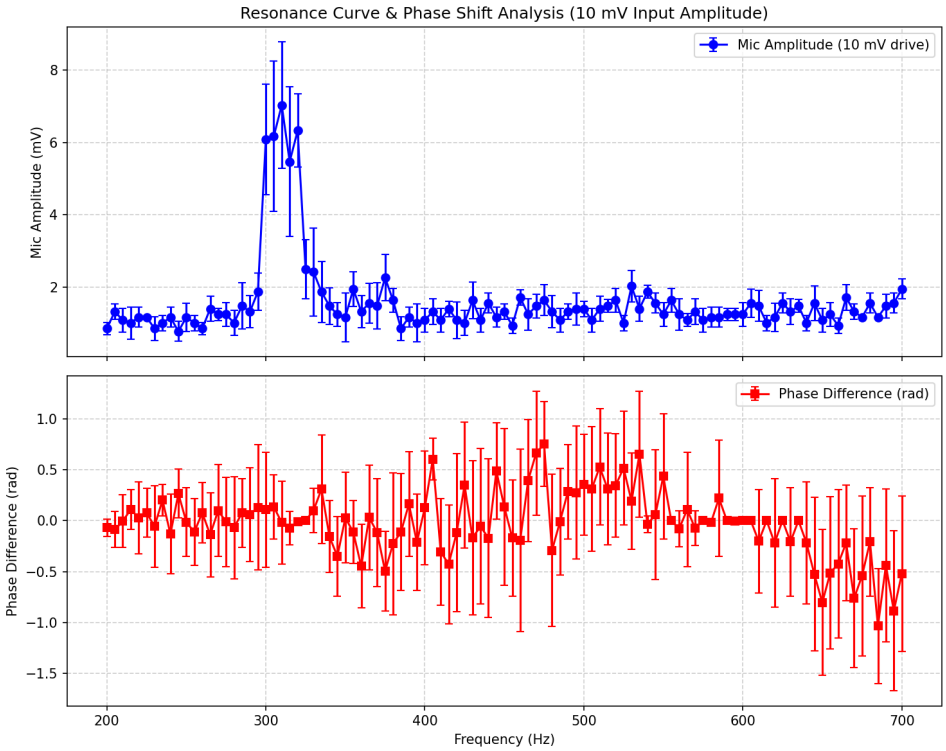


Figure 1: **Acoustic Frequency Response and Phase Shift at 10 mV Drive Amplitude.** (**Top**) Microphone response amplitude (mV) showing a clear fundamental resonance peak centered near 310 Hz. (**Bottom**) Measured phase difference (rad) between the microphone signal and the reference drive signal across the 200–700 Hz sweep. Error bars indicate standard deviation over repeated captures per frequency step.

Plan for Upcoming Weeks

- We will focus on completing the first part of the experiment, including the experimental measurements and analysis of the resonance characteristics of the soda-can Helmholtz resonator.
- Then in the next upcoming week, We will try to construct and characterize an acoustic antenna using the arrays of soda cans and also try to study the directivity and pattern of the antenna.
- Subject to the availability of time, we will try to investigate the dependence of the resonance frequency on the opening area of the can by varying its opening size (it's not shown in the paper but we can do).